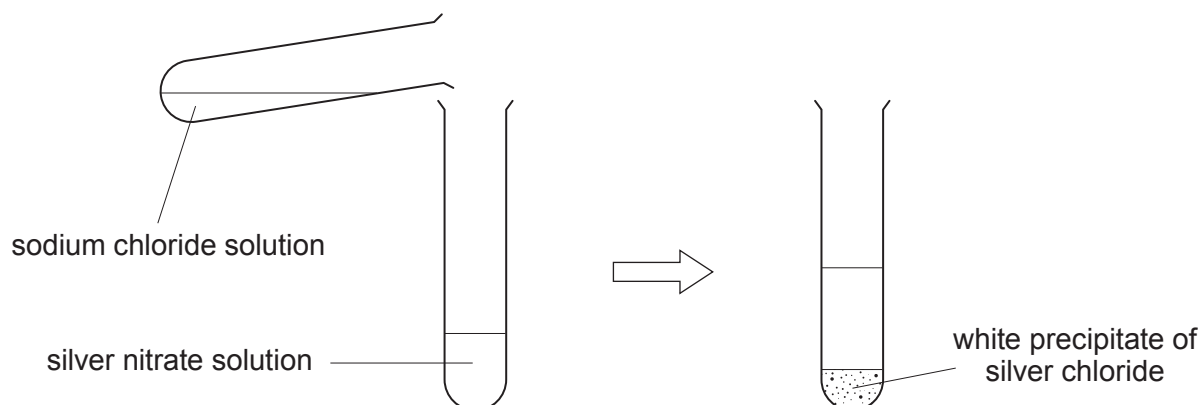


5. (a) Dafydd was asked to make some silver chloride. He formed a white precipitate of silver chloride by mixing solutions of sodium chloride and silver nitrate.



- (i) Complete the symbol equation for this reaction. [1]



- (ii) Put a tick (✓) in the box next to the statement which describes why this method works. [1]

silver is more dense than sodium

☐

silver chloride is soluble

☐

silver chloride is insoluble

☐

silver is below sodium in the reactivity series

☐

- (iii) Give the name of the process that you would use to separate the precipitate of silver chloride from the reaction mixture. [1]

.....



- (b) Calculate the relative formula mass (M_r) of silver nitrate, AgNO_3 . [2]

$$A_r(\text{O}) = 16$$

$$A_r(\text{N}) = 14$$

$$A_r(\text{Ag}) = 108$$

$$M_r = \dots\dots\dots$$

- (c) The relative formula mass (M_r) of sodium chloride, NaCl , is 58.5.

Calculate the percentage of sodium in sodium chloride. Give your answer to **1 decimal place**. [2]

$$A_r(\text{Na}) = 23$$

$$\text{Percentage} = \dots\dots\dots \%$$

